THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 8–11: **Introduction to Trigonometry, Some Applications of Trigonometry, Circles, Areas Related to Circles** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** $\frac{4}{5}$	Hypotenuse is 5
2	**(b)** $15\sqrt{3}$ m	$h = 15\tan 60^\circ = 15 \times \sqrt{3}$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height $= 15\sqrt{3}$ m
3	**(a)** 50°	$\angle AOB = 100^\circ$ and OP bisects it
4	**(a)** ₹ 880	Cost $= 88 \times 10$; Cost = ₹ 880
5	**(a)** $\frac{12}{5}$	Find the perpendicular
6	**(c)** $3\sqrt{3}$ m	Height $= 6\sin 60^\circ = 6 \times \frac{\sqrt{3}}{2}$; Height reached on the wall $= 3\sqrt{3}$ m
7	**(d)** 30 cm	$PT^2 = OP^2 - OT^2 = 34^2 - 16^2 = 900$; $PT = 30$ cm
8	**(d)** 50	Distance 66 m = 6600 cm; revolutions $= \frac{6600}{132}$; Revolutions $= 50$
9	**(c)** 1	Substitute the standard values
10	**(d)** 12 m	Add the height of the eyes: $2 + 10$; Height of building = 12 m
11	**(c)** 5 cm	$9 + 7 = AD + 11$; $AD = 5$ cm
12	**(d)** 28 cm	$R = r_1 + r_2 = 19 + 9$; $R = 28$ cm
13	**(b)** 1	$\tan(90^\circ - \theta) = \cot\theta$
14	**(b)** $10\sqrt{3}$ m	$h(\cot 30^\circ - \cot 60^\circ) = 20$ gives $h = 20 \div (\frac{2\sqrt{3}}{3})$; Height $= 10\sqrt{3}$ m (distances 30 m and 10 m)
15	**(a)** 30°	$\angle OAP = 90^\circ$ (radius ⊥ tangent); $\angle OAB = 90^\circ - 60^\circ = 30^\circ$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Hypotenuse = 5 (1); $\sec A = \frac{5}{3}$ (1).	1+1
17	Broken part (hypotenuse) $= \frac{6}{\sin 30^\circ} = 12$ m; Distance from the foot $= 6\cot 30^\circ = 6\sqrt{3}$ m (1); Distance from the foot of the tree to the point where its top touches the ground $= 6\sqrt{3}$ m (1).	1+1
18	Tangents from an external point are equal: $QA = QB$; $3x + 4 = 2x + 9$ gives $x = 5$ (1); $QA = 3(5) + 4 = 19$ cm (1).	1+1
19	Angle turned in 15 minutes $= 90^\circ$; Arc $= \frac{90}{360} \times 2 \times \frac{22}{7} \times 14$ (1); Distance = 22 cm (1).	1+1
20	$\cos 65^\circ = \sin 25^\circ$ and $\cot 50^\circ = \tan 40^\circ$ (2); value $= 1 + 1 = 2$ (1).	2+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Hypotenuse $= \sqrt{12^2 + 5^2} = 13$ (1); $\sin\theta = \frac{12}{13}$, $\cos\theta = \frac{5}{13}$ (1); $\sec\theta = \frac{13}{5}$ (1).	1+1+1
22	Let height $= h$; the distances from the foot are $h\cot 30^\circ$ and $h\cot 60^\circ$, which differ by 20 m (1); $h(\cot 30^\circ - \cot 60^\circ) = 20$ gives $h = 20 \div (\frac{2\sqrt{3}}{3})$ (1); Height $= 10\sqrt{3}$ m (distances 30 m and 10 m) (1).	1+1+1
23	Tangent lengths from a vertex are equal: $AF = AE = x$, $BD = BF = y$, $CD = CE = z$ (1); $x + y = 12$, $y + z = 7$, $z + x = 9$; adding, $x + y + z = 14$ (1); $AF = 7$ cm, $BD = 5$ cm, $CE = 2$ cm (1).	1+1+1
24	Segment = sector − triangle (1); Sector $= 38.5$ cm^2, triangle $= \frac{1}{2} \times 7 \times 7 = 24.5$ cm^2 (1); Segment $= 38.5 - 24.5 = 14$ cm^2 (1).	1+1+1
	OR: $AC = \sqrt{7^2 + 24^2} = 25$ cm (1); $\sin A = \frac{BC}{AC} = \frac{24}{25}$, $\cos A = \frac{AB}{AC} = \frac{7}{25}$ (1); $\tan A = \frac{BC}{AB} = \frac{24}{7}$ (1).	1+1+1
25	Horizontal distance $d = 5\cot 45^\circ = 5$ m (from the depression of the foot) (1); Part of the tower above the top of the building $= d\tan 60^\circ = 5\sqrt{3}$ m (1); Tower $= 5 + 5\sqrt{3}$ m (1).	1+1+1
	OR: $AC = 29$ cm; the incircle touches AB, BC at F, D and $BDOF$ is a square of side r (1); $AF = AB - r = 20 - r$, $CD = BC - r = 21 - r$ and $AF + CD = AC$ (tangent lengths) (1); $(20 - r) + (21 - r) = 29$ gives $r = 6$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

26	LHS $= \dfrac{\sin^2\theta + (1+\cos\theta)^2}{\sin\theta(1+\cos\theta)}$ (1); $= \dfrac{\sin^2\theta + 1 + 2\cos\theta + \cos^2\theta}{\sin\theta(1+\cos\theta)} = \dfrac{2+2\cos\theta}{\sin\theta(1+\cos\theta)}$ (2); $= \dfrac{2}{\sin\theta} = 2\csc\theta$ = RHS (2).	5
	OR: Let height $= h$; distances $h\cot 60^\circ$ and $h\cot 30^\circ$ add up to 80 (2); $h\left(\frac{1}{\sqrt{3}} + \sqrt{3}\right) = 80$ gives $h = 20\sqrt{3}$ (2); Height $20\sqrt{3}$ m; distances 20 m and 60 m (1).	2+2+1
27	Let the tangent XY touch the circle with centre O at P, and take any other point Q on XY (1); Q lies outside the circle (a tangent meets the circle only at P), so $OQ > OP$ (1); This holds for every point $Q \neq P$ on XY, so OP is the shortest distance from O to the line XY (2); The shortest distance is the perpendicular, so $OP \perp XY$ (1).	1+1+2+1
	OR: Segment $=$ sector $-$ triangle (2); Sector $= 102.67$ cm^2; triangle $= \frac{\sqrt{3}}{4}r^2 = 84.77$ cm^2 (2); Segment $= 102.67 - 84.77 = 17.90$ cm^2 (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 8–11: **Introduction to Trigonometry, Some Applications of Trigonometry, Circles, Areas Related to Circles** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 10 cm	$R^2 = 6^2 + 8^2 = 100$; $R = 10$ cm
2	**(a)** 1	$\cos(90^\circ - \theta) = \sin\theta$
3	**(c)** 12 m	Distance from the foot $= 4\cot 30^\circ = 4\sqrt{3}$ m; Original height of the tree $= 12$ m (that is $4 + 8$)
4	**(a)** 41 cm	$OP^2 = r^2 + PT^2 = 9^2 + 40^2 = 1681$; $OP = 41$ cm
5	**(a)** 115.5 cm²	Area $= \frac{30}{360} \times \frac{22}{7} \times 21^2$; Area $= 115.5$ cm^2
6	**(a)** 4	$k^2(\sin^2 + \cos^2)$
7	**(a)** $7 + 7\sqrt{3}$ m	Part of the tower above the top of the building $= d\tan 60^\circ = 7\sqrt{3}$ m; Tower $= 7 + 7\sqrt{3}$ m
8	**(c)** 2 cm	$AF = AB - r = 6 - r$, $CD = BC - r = 8 - r$ and $AF + CD = AC$ (tangent lengths); $(6 - r) + (8 - r) = 10$ gives $r = 2$ cm
9	**(a)** 38.5 m²	Area $= \frac{90}{360} \times \frac{22}{7} \times 7^2$; Area $= 38.5$ m^2
10	**(b)** $\frac{3}{5}$	Reciprocal
11	**(d)** 45°	$\tan\theta = 1$ which is $\tan 45^\circ$; $\theta = 45^\circ$
12	**(c)** 17 cm	$3x + 5 = 2x + 9$ gives $x = 4$; $QA = 3(4) + 5 = 17$ cm
13	**(c)** 60°	$\theta = \frac{231 \times 360}{\frac{22}{7} \times 21^2}$; $\theta = 60^\circ$
14	**(d)** $\frac{12}{13}$	Hypotenuse is 13
15	**(c)** $7\sqrt{3}$ m	$h = 21\tan 30^\circ = 21 \times \frac{\sqrt{3}}{3}$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height $= 7\sqrt{3}$ m

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	The ladder is the hypotenuse, length 14 m; Base $= 14\cos 60^\circ = 14 \times \frac{1}{2}$ (1); Distance of the foot of the ladder from the wall $= 7$ m (1).	1+1
17	$PA = PB$, so $\angle PAB = \frac{180^\circ - 40^\circ}{2} = 70^\circ$; $\angle OAP = 90^\circ$ (radius ⊥ tangent) (1); $\angle OAB = 90^\circ - 70^\circ = 20^\circ$ (1).	1+1
18	Perimeter of the square = circumference $= 2\pi r = 88$ cm; Side $= 88 \div 4 = 22$ cm (1); Area $= 22^2 = 484$ cm^2 (1).	1+1
19	$\cot 69^\circ = \tan 21^\circ$ (1); value $= 1$ (1).	1+1
20	$\tan\theta = \frac{\text{rise}}{\text{run}} = \frac{10\sqrt{3}}{10}$; $\tan\theta = \sqrt{3}$ which is $\tan 60^\circ$ (1); $\theta = 60^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Perimeter = arc + $2r$ (1); Arc $= \frac{90}{360} \times 2\pi \times 14 = 22$ cm (1); Perimeter $= 22 + 28 = 50$ cm (1).	1+1+1
22	$(1 - \cos^2\theta) = \sin^2\theta$ (1); $\sin^2\theta \cdot \csc^2\theta$ (1) $= 1$ (1).	1+1+1
23	Distances from the foot: $h\cot 30^\circ = 75\sqrt{3}$ m (farther) and $h\cot 45^\circ = 75$ m (nearer) (1); Both objects are on the same side, so the distances are subtracted (1); Distance $= 75\sqrt{3} - 75$ m (1).	1+1+1
24	The chord touches the smaller circle at M, so $OM \perp$ chord and M is the midpoint of the chord (1); Half-chord $= \sqrt{5^2 - 3^2} = 4$ cm (1); Chord $= 2 \times 4 = 8$ cm (1).	1+1+1
	OR: The 8 ribs divide the circle into 8 equal sectors (1); Area of the circle $= \frac{22}{7} \times 28^2 = 2464$ cm^2 (1); Area between ribs $= 2464 \div 8 = 308$ cm^2 (1).	1+1+1
25	Hypotenuse $= \sqrt{15^2 + 8^2} = 17$ (1); $\sin\theta = \frac{15}{17}$, $\cos\theta = \frac{8}{17}$ (1); $\sec\theta = \frac{17}{8}$ (1).	1+1+1
	OR: Let height $= h$; the distances from the foot are $h\cot 30^\circ$ and $h\cot 60^\circ$, which differ by 34 m (1); $h(\cot 30^\circ - \cot 60^\circ) = 34$ gives $h = 34 \div (\frac{2\sqrt{3}}{3})$ (1); Height $= 17\sqrt{3}$ m (distances 51 m and 17 m) (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Divide numerator and denominator by $\cos A$ (1): $\dfrac{4\tan A+3}{4\tan A+3}$ (1); $=\dfrac{4(\frac{4}{3})+3}{4(\frac{4}{3})+3}=1$ (2); verified with $\sin A=\frac{4}{5}$, $\cos A=\frac{3}{5}$ (1).	1+1+2+1
	OR: Let pedestal $=p$; distance from the point $=p\cot 45^\circ=p$ (2); $p+2=p\tan 60^\circ$ gives $p(\sqrt{3}-1)=2$ (2); $p=\frac{2}{\sqrt{3}-1}=1+\sqrt{3}$ m (1).	2+2+1
27	$AC=\sqrt{6^2+8^2}=10$ cm; $BDOF$ is a square of side r, so $BF=BD=r$ (2); $AF=AE=6-r$ and $CD=CE=8-r$ and $AE+EC=AC$: $(6-r)+(8-r)=10$ (2); $r=2$ cm, $AF=4$ cm, $CD=6$ cm (1).	2+2+1
	OR: Wire $=$ circumference $+$ the diameters (2); Circumference $=\pi d=110$ mm; diameters $=5\times 35=175$ mm (2); Total $=285$ mm (1).	2+2+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 8–11: **Introduction to Trigonometry, Some Applications of Trigonometry, Circles, Areas Related to Circles** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** $\sqrt{119}$ cm	$PQ^2 = 144 - 25 = 119$
2	**(c)** ₹ 660	Cost $= 132 \times 5$; Cost = ₹ 660
3	**(d)** $\frac{15}{8}$	Find the perpendicular
4	**(c)** $8\sqrt{3}$ m	Height $= 16 \sin 60^\circ = 16 \times \frac{\sqrt{3}}{2}$; Height reached on the wall $= 8\sqrt{3}$ m
5	**(d)** 28 cm	$PT^2 = OP^2 - OT^2 = 35^2 - 21^2 = 784$; $PT = 28$ cm
6	**(a)** 50	Distance 110 m = 11000 cm; revolutions $= \frac{11000}{220}$; Revolutions $= 50$
7	**(d)** $\frac{8}{3}$	Substitute the standard values
8	**(b)** 18 m	Add the height of the eyes: $3 + 15$; Height of building = 18 m
9	**(b)** 10 cm	$12 + 8 = AD + 10$; $AD = 10$ cm
10	**(d)** 14 cm	$R = r_1 + r_2 = 8 + 6$; $R = 14$ cm
11	**(a)** 1	$\tan(90^\circ - \theta) = \cot\theta$
12	**(d)** $15\sqrt{3}$ m	$h(\cot 30^\circ - \cot 60^\circ) = 30$ gives $h = 30 \div (\frac{2\sqrt{3}}{3})$; Height $= 15\sqrt{3}$ m (distances 45 m and 15 m)
13	**(c)** 35°	$\angle OAP = 90^\circ$ (radius ⊥ tangent); $\angle OAB = 90^\circ - 55^\circ = 35^\circ$
14	**(a)** 13 cm	$R^2 = 5^2 + 12^2 = 169$; $R = 13$ cm
15	**(d)** 1	$\cos(90^\circ - \theta) = \sin\theta$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	For a quadrilateral circumscribed about a circle, $AB + CD = AD + BC$; $13 + 8 = AD + 9$ (1); $AD = 12$ cm (1).	1+1
17	Grazed area is a quadrant: $\frac{1}{4}\pi r^2$; Increase $= \frac{1}{4} \times \frac{22}{7}(14^2 - 7^2)$ (1); Increase = 115.5 m² (1).	1+1
18	Hypotenuse = 13 (1); $\sec A = \frac{13}{5}$ (1).	1+1
19	Broken part (hypotenuse) $= \frac{8}{\sin 30^\circ} = 16$ m; Distance from the foot $= 8 \cot 30^\circ = 8\sqrt{3}$ m (1); Distance from the foot of the tree to the point where its top touches the ground $= 8\sqrt{3}$ m (1).	1+1
20	Tangents from an external point are equal: $QA = QB$; $5x + 1 = 3x + 9$ gives $x = 4$ (1); $QA = 5(4) + 1 = 21$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Tangent lengths from a vertex are equal: $AF = AE = x$, $BD = BF = y$, $CD = CE = z$ (1); $x + y = 14$, $y + z = 10$, $z + x = 12$; adding, $x + y + z = 18$ (1); $AF = 8$ cm, $BD = 6$ cm, $CE = 4$ cm (1).	1+1+1
22	Segment = sector − triangle (1); Sector = 154 cm², triangle $= \frac{1}{2} \times 14 \times 14 = 98$ cm² (1); Segment $= 154 - 98 = 56$ cm² (1).	1+1+1
23	$AC = \sqrt{20^2 + 21^2} = 29$ cm (1); $\sin A = \frac{BC}{AC} = \frac{21}{29}$, $\cos A = \frac{AB}{AC} = \frac{20}{29}$ (1); $\tan A = \frac{BC}{AB} = \frac{21}{20}$ (1).	1+1+1
24	Horizontal distance $d = 8 \cot 45^\circ = 8$ m (from the depression of the foot) (1); Part of the tower above the top of the building $= d \tan 60^\circ = 8\sqrt{3}$ m (1); Tower $= 8 + 8\sqrt{3}$ m (1).	1+1+1
	OR: $AC = 40$ cm; the incircle touches AB, BC at F, D and $BDOF$ is a square of side r (1); $AF = AB - r = 24 - r$, $CD = BC - r = 32 - r$ and $AF + CD = AC$ (tangent lengths) (1); $(24 - r) + (32 - r) = 40$ gives $r = 8$ cm (1).	1+1+1
25	Perimeter = arc + $2r$ (1); Arc $= \frac{180}{360} \times 2\pi \times 14 = 44$ cm (1); Perimeter $= 44 + 28 = 72$ cm (1).	1+1+1
	OR: LHS $= \frac{\sin^2\theta + \cos^2\theta}{\sin\theta\cos\theta}$ (2); $= \frac{1}{\sin\theta\cos\theta} = \sec\theta\csc\theta$ (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

26	LHS $= \left(\frac{1-\cos\theta}{\sin\theta}\right)^2$ (2); $= \frac{(1-\cos\theta)^2}{1-\cos^2\theta}$ (1); $= \frac{(1-\cos\theta)^2}{(1-\cos\theta)(1+\cos\theta)} = \frac{1-\cos\theta}{1+\cos\theta}$ (2).	5
	OR: Let height $= h$; distances $h\cot 60^\circ$ and $h\cot 30^\circ$ add up to 120 (2); $h\left(\frac{1}{\sqrt{3}} + \sqrt{3}\right) = 120$ gives $h = 30\sqrt{3}$ (2); Height $30\sqrt{3}$ m; distances 30 m and 90 m (1).	2+2+1
27	Let PA and PB be tangents from P to the circle with centre O; join OA, OB, OP (1); $\angle OAP = \angle OBP = 90^\circ$ (radius $\perp$ tangent) (1); In $\triangle OAP$ and $\triangle OBP$: $OA = OB$ (radii), OP common, right angles equal (1); $\triangle OAP \cong \triangle OBP$ (RHS) (1); Hence $PA = PB$ (CPCT) (1).	1+1+1+1+1
	OR: Segment $=$ sector $-$ triangle (2); Sector $= 231.00$ cm^2; triangle $= \frac{\sqrt{3}}{4}r^2 = 190.73$ cm^2 (2); Segment $= 231.00 - 190.73 = 40.27$ cm^2 (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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